Neerav Chaudhary

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LinkedIn | GitHub | Portfolio

APPLIED AI / PRODUCT ENGINEER | RESEARCH, EVALUATION, AI SAFETY, PRODUCT DESIGN

Applied AI / Product Engineer and published researcher who builds complete AI products, from user workflows and interfaces to safety protocols and evaluation. Worked with energy-sector response teams on AI/ML compliance systems and with founders and policymakers on AI-safety research.

AI PRODUCTS AND SYSTEMS

Relay - Employee operations, conversational AI | React, TypeScript, Fastify, Deepgram **Sep-Oct 2026**

- Cut employee onboarding and support workflow steps by 67%, consolidating record lookup, team handoff, and action confirmation into one voice AI workflow that automated approvals and claims across People Ops, IT, Finance, and Administration.
- Enabled employees to complete multi-step operations requests entirely by voice, from record lookup to confirmed completion, without switching channels (2.0-second median response latency; 51 ms p95 interruption handling).
- Completed 93.6% of development workflows and 92.9% of held-out request variants across 252 conversations and 764 turns. Relay reached the correct record in 99.5% of tool runs, blocked all 30 fabricated IDs, and made zero unsafe state changes across 21 adversarial runs.

TraceLens - Health-chatbot safety evaluation and decision support | Python, PyTorch, Hugging Face **Oct-Nov 2026**

- Ran an independent expert safety model alongside 11 health chatbots to improve emergency detection, evaluating 2,970 responses to measure when it caught missed emergencies and corrected final guidance.
- Delivered evidence-backed emergency guidance in 0.35 seconds by identifying risk, recommending the next action, and abstaining when evidence was insufficient (99.5% faster than the 73.2-second median chatbot response).
- Caught and disclosed 18 of 31 emergencies missed by health chatbots, increasing correct final guidance from 39/70 to 47/70 while adding only one false alarm across 134 non-emergency cases (held-out AUC 0.966).

PROFESSIONAL EXPERIENCE

NTPC | AI/ML Engineer Intern **Nov 2023-May 2024**

- Identified the exact source of pollution behind severe PM2.5 spikes - plant-caused or regional - with 95% attribution accuracy using XGBoost + SHAP, giving the compliance team a clear basis for incident assessment and response. (Machine Learning, Python)
- Forecasted PM2.5 limit breaches 24 hours in advance so operators could curtail emissions proactively, while a virtual sensor filled monitoring gaps when air sensors were offline for maintenance, raising regulatory data coverage from 92% to 98%.
- Cut pollution incident-to-report turnaround from approximately nine hours to two by automating compliance analysis and report generation. A local open-source LLM turned monitoring and source-attribution results into internal briefs and regulator letters.
- Guided monitoring-equipment procurement by turning plant and regulatory (CPCB) requirements into clear criteria for choosing between high-end reference stations and low-cost sensor networks, avoiding overspend and compliance gaps.

Deep Strat | AI Safety and Research Intern **Aug-Nov 2023**

- Built and evaluated a three-stage LLM pipeline for intent and sentiment classification, automated team routing, and SOP-grounded report generation under a research grant demonstrating LLM effectiveness in business workflows. Designed the evaluation methodology (routing accuracy and report-quality criteria) and embedded human-in-the-loop guardrails.
- Led a comparative evaluation of churn-prediction models, benchmarking Random Forest against CNN across 2,000 records with a cost-sensitive objective; tuned the decision threshold to cut false alerts by 25% at constant AUC, sharpening retention outreach toward higher-risk customers and producing a reusable model-evaluation framework.
- Authored a 50-page AI-safety reference on model failure modes and guardrails for founders, the COO, and policy researchers, informing jointly authored research presented at Nullcon and an internal curriculum spanning classical ML through RLHF.

Manipal University Jaipur | Medical Imaging Researcher **Aug-Dec 2022**

- Improved reported brain-atrophy analysis accuracy from 66% to 71% by focusing on the hippocampal region across 2,198 ADNI MRI images. Built the interpretable Python/OpenCV pipeline and published the results as first author at IEEE ICRITO.
- Improved held-out four-class MRI classification accuracy from 78% to 85% across 6,400 images using probability-level ensembling in TensorFlow.

PUBLICATION

IEEE ICRITO 2022 | First Author, Computer Vision Research

First-author study of dementia-related brain atrophy across 2,198 ADNI MRI images; hippocampal analysis improved reported accuracy from 66% to 71%.

doi:10.1109/ICRITO56286.2022.9964994

EDUCATION

San Jose State University | M.S., Industrial and Engineering Management | GPA 3.81 **2024-2026**

Manipal University Jaipur | B.Tech., Computer Engineering | GPA 8.8/10 **2018-2022**

SKILLS

AI products and safety: end-to-end product design, conversational AI, voice agents, workflow automation, LLM evaluation, evaluation harness engineering, fine-tuning, structured tools, guardrails, RAG, Deepgram, Hugging Face

Product engineering: interaction design, UI/UX, Python, TypeScript, React, FastAPI, Fastify, SQL, PostgreSQL, SQLite, REST APIs, Docker, AWS, Git

ML and research: XGBoost, SHAP, LSTM, Random Forest, PyTorch, TensorFlow, scikit-learn, OpenCV, forecasting, attribution, calibration